Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 08

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Activity 01

bisheka@bisheka-VirtualBox: ~

GNU nano 7.2 Lab8Act1.c

```c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main() {
     key_t  key=1234;
     int shmid;
     char *ptr;

   shmid=shmget(key,4096,0666|IPC_CREAT);

   ptr=(char*)shmat(shmid,NULL,0);

  sprintf(ptr,"Hello from shared memory");

   printf("Data written to shared  memory\n");

    shmdt(ptr);
   return 0;
}
```

bisheka@bisheka-VirtualBox: ~

```
bisheka@bisheka-VirtualBox:~$ nano Lab8Act1.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab8Act1 Lab8Act1.c
bisheka@bisheka-VirtualBox:~$ ./Lab8Act1
Data written to shared  memory
bisheka@bisheka-VirtualBox:~$
```

#include <stdio.h>

#include <sys/ipc.h>

#include <sys/shm.h>

#include <string.h>

- stdio.h - input/output functions
- sys/ipc.h - IPC (Inter Process Communication)
- sys/shm.h - shared memory functions
- string.h - string handling

- **key_t key = 1234;** - Unique key to identify shared memory segment
- **int shmid;** - shared memory ID
- **char *ptr;** - pointer to access memory

shmid = shmget(key, 4096, 0666 | IPC_CREAT);

- key - identifier
- 4096 - size (bytes)
- 0666 - read/write permissions
- IPC_CREAT - create if not exists

ptr = (char *) shmat(shmid, NULL, 0); - Attaches shared memory to process, Returns pointer to memory

sprintf(ptr, "Hello from shared memory!"); - Writes data into shared memory

printf("Data written to shared memory\n"); - Displays confirmation message

shmdt(ptr); - Detaches shared memory from process

Activity 02

bisheka@bisheka-VirtualBox: ~

GNU nano 7.2 Lab8Act2.c

```c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
        key_t key=1234;
        int shmid;
        char *ptr;

        shmid = shmget(key,4096,0666);

        ptr=(char*)shmat(shmid,NULL,0);

        printf("Read from shared memory:%s\n",ptr);

         shmdt(ptr);

         shmctl(shmid,IPC_RMID,NULL);

         return 0;
}
```

bisheka@bisheka-VirtualBox: ~

```
bisheka@bisheka-VirtualBox:~$ nano Lab8Act2.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab8Act2 Lab8Act2.c
bisheka@bisheka-VirtualBox:~$ ./Lab8Act2
Read from shared memory:Hello from shared memory
bisheka@bisheka-VirtualBox:~$
```

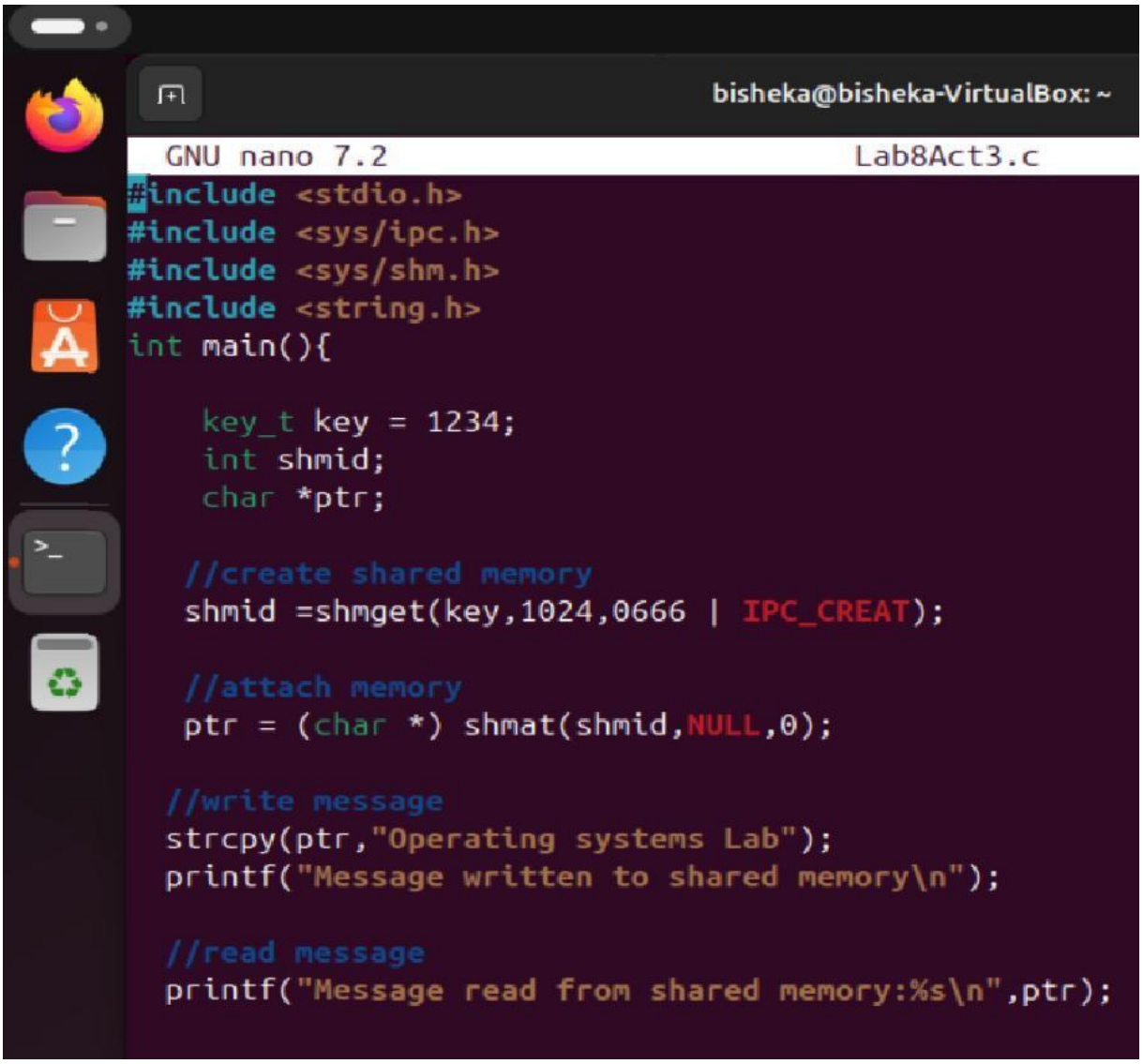
shmid = shmget(key, 4096, 0666); - Access existing shared memory (no IPC_CREAT)

ptr = (char *) shmat(shmid, NULL, 0); - Attach memory

printf("Data: %s\n", ptr); - Read and print data

shmdt(ptr); - Detach memory

Activity 03



bisheka@bisheka-VirtualBox: ~

GNU nano 7.2 Lab8Act3.c

```
    char *ptr;

   //create shared memory
   shmid =shmget(key,1024,0666 | IPC_CREAT);

   //attach memory
   ptr = (char *) shmat(shmid,NULL,0);

  //write message
  strcpy(ptr,"Operating systems Lab");
  printf("Message written to shared memory\n");

  //read message
  printf("Message read from shared memory:%s\n",ptr);

  //detach memory
  shmdt(ptr);

  //delete shared memory
  shmctl(shmid,IPC_RMID,NULL);
  return 0;
}
```

bisheka@bisheka-VirtualBox: ~

```
bisheka@bisheka-VirtualBox:~$ nano Lab8Act3.c
bisheka@bisheka-VirtualBox:~$ gcc -oLab8Act3 Lab8Act3.c
bisheka@bisheka-VirtualBox:~$ ./Lab8Act3
Message written to shared memory
Message read from shared memory:Operating systems Lab
bisheka@bisheka-VirtualBox:~$
```